

Johanna Loyer

Postdoctoral researcher in Cryptology

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Research interests

My research focuses on the cryptanalysis of security **lattice-based** and **code-based** schemes, in both classical and **quantum** settings. I am particularly interested in transferring techniques between these domains.

Work Experience

febr. 2025 – sept. 2026 **Post-doctoral researcher**, *Inria Saclay*, GRACE team.

Supervisor : Thomas Debris-Alazard

febr. 2024 – jan. 2025 **Post-doctoral researcher**, *CWI*, Amsterdam, Cryptology group.

Supervisor : Léo Ducas

sept. 2020 – dec. 2023 **Ph.D.**, *Inria*, Paris, COSMIQ team.

Thesis: “*Quantum Cryptanalysis of Lattices and Codes*”

Advisors: André Chailloux and Nicolas Sendrier

sept. 2020 – jan. 2023 **Teaching assistant** (128h), *Sorbonne University*, Paris.

Quantum circuits and logic gates, Master 1

Quantum information, Master 1

Programming in Python, Bachelor 1

Education

Sept. 2018 – Aug. 2020 **Master in Applied Mathematics**, *University of Limoges, France*.

March – August 2020 **Master's thesis internship**, *Inria Paris*

Supervisor: André Chailloux

Sept. 2017 – Aug. 2018 1st year in **Engineering School in Cybersecurity**, *INSA Centre Val de Loire*, Bourges.

Research Publications

CRYPTO 2025 **Wagner's Algorithm Provably Runs in Subexponential Time for SIS^∞** , with Léo Ducas and Lynn Engelberts.

CIC 2025 **Lattice Reduction via Dense Sublattices: A Cryptanalytic No-Go**
with Léo Ducas

DCC 2024 **Quantum sieving for code-based cryptanalysis and its limitations for ISD**,
with Lynn Engelberts and Simona Etinski.

PQCrypto 2023 **Classical and quantum 3 and 4-sieves to solve SVP with low memory**, with André Chailloux.

ASIACRYPT 2021 **Lattice sieving via quantum random walks**, with André Chailloux.

Preprints

A Tight Quantum Algorithm for Multiple Collision Search, avec Xavier Bonnetain, André Schrottenloher et Yixin Shen.
Quantum security analysis of Wave.

Other contributions

Comité de lecture

I am in the program committee of PKC 2026. I have been a sub-reviewer for Eurocrypt 2023, ANTS 2024, Crypto 2025 and Eurocrypt 2025.

Participation in standardisation effort

NIST Post-Quantum **Wave digital signature scheme**, with Gustavo Banegas, Kévin Carrier, André Cryptography 2023, Chailloux, Alain Couvreur, Thomas Debris-Alazard, Philippe Gaborit, Pierre Round 1 Karpman, Ruben Niederhagen, Nicolas Sendrier, Benjamin Smith, Jean-Pierre Tillich.

Outreach and Science Communication

2024 Interviewed by a science journalist for *Epsilon*, a science popularization magazine.

2023 Organized an outreach event to present my Ph.D. research and raise awareness about cybersecurity and privacy issues.

2021 Participated in the RJMI events to promote STEM careers among high school girls.

2020 Talks in middle schools: *"Introduction to cryptology: RSA"*.

2018 – 2019 Talks in high schools: *"How mathematics are useful in daily life?"* in response to a national reform that removed mathematics as a mandatory subject.

2018 Robotics and programming workshops in elementary schools, in partnership with the Graduate School of Teaching and Education (ESPE) of Bourges.

